

CASE REPORT

Spontaneous regression of recurrent pulmonary large cell neuroendocrine carcinoma with alteration of PD-L1 expression after surgical resection: A case report

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Abstract

Pulmonary large cell neuroendocrine carcinoma (LCNEC) is a rare and aggressive subtype of non-small cell lung cancer with a poor prognosis. Spontaneous regression, that is, partial or complete disappearance of a malignancy without medical intervention, is extremely rare in LCNEC. Herein, we present a case of spontaneous complete regression in a 71-year-old male patient with recurrent LCNEC after surgical resection. The patient was diagnosed with stage IB LCNEC and underwent surgical resection. At 1-year follow-up, chest computed tomography revealed a recurrent lesion next to the stump site and enlargement of lymph nodes 4R and 7; recurrent LCNEC was confirmed. The patient declined chemoradiation therapy. One year after recurrence, the patient experienced severe multifocal necrotizing pneumonia and was treated with antibiotics, resulting in a gradual decrease in the size of the recurrent lesion. Five years after the initial diagnosis, positron emission tomography/computed tomography revealed no hypermetabolic lesions, indicating the spontaneous complete regression of LCNEC.

KEYWORDS

lung cancer, pulmonary large-cell neuroendocrine carcinoma, spontaneous regression

INTRODUCTION

Large cell neuroendocrine carcinoma (LCNEC) is a rare form of primary lung cancer with a low incidence rate of approximately 1.35 cases per 100 000 individuals per year.¹ Although LCNEC is classified as a subtype of non-small cell lung cancer (NSCLC), it possesses unique traits that make it more aggressive than other types of NSCLC. Its biological behavior closely resembles that of small cell lung cancer.^{2,3} Spontaneous regression of malignancy refers to the partial or complete disappearance of a malignant disease without the use of medical intervention.⁴ However, such regression is extremely rare in patients with NSCLC, especially LCNEC.⁵ Here, we report a case of spontaneous complete regression of LCNEC in a patient with recurrent lymph node metastasis.

CASE REPORT

A 71-year-old man was admitted to our hospital's outpatient clinic with a lung mass detected on chest computed tomography (CT) which had been performed at another hospital. He had a smoking history of 30 pack-years but no complaints of respiratory symptoms. Initial chest CT revealed a 26 × 21-mm lobulated soft-tissue nodule with a multilobulated margin in the right middle lobe (RML) (Figure 1a,b). Endobronchial ultrasound-guided biopsy of the lung nodule was performed. Histological examination revealed that it was an undetermined malignant tumor. Positron emission tomography/computed tomography (PET-CT) revealed no metastases (Figure 1c). RML lobectomy was performed via video-assisted thoracoscopic surgery. Pathological examination of the excised specimen revealed that it was a

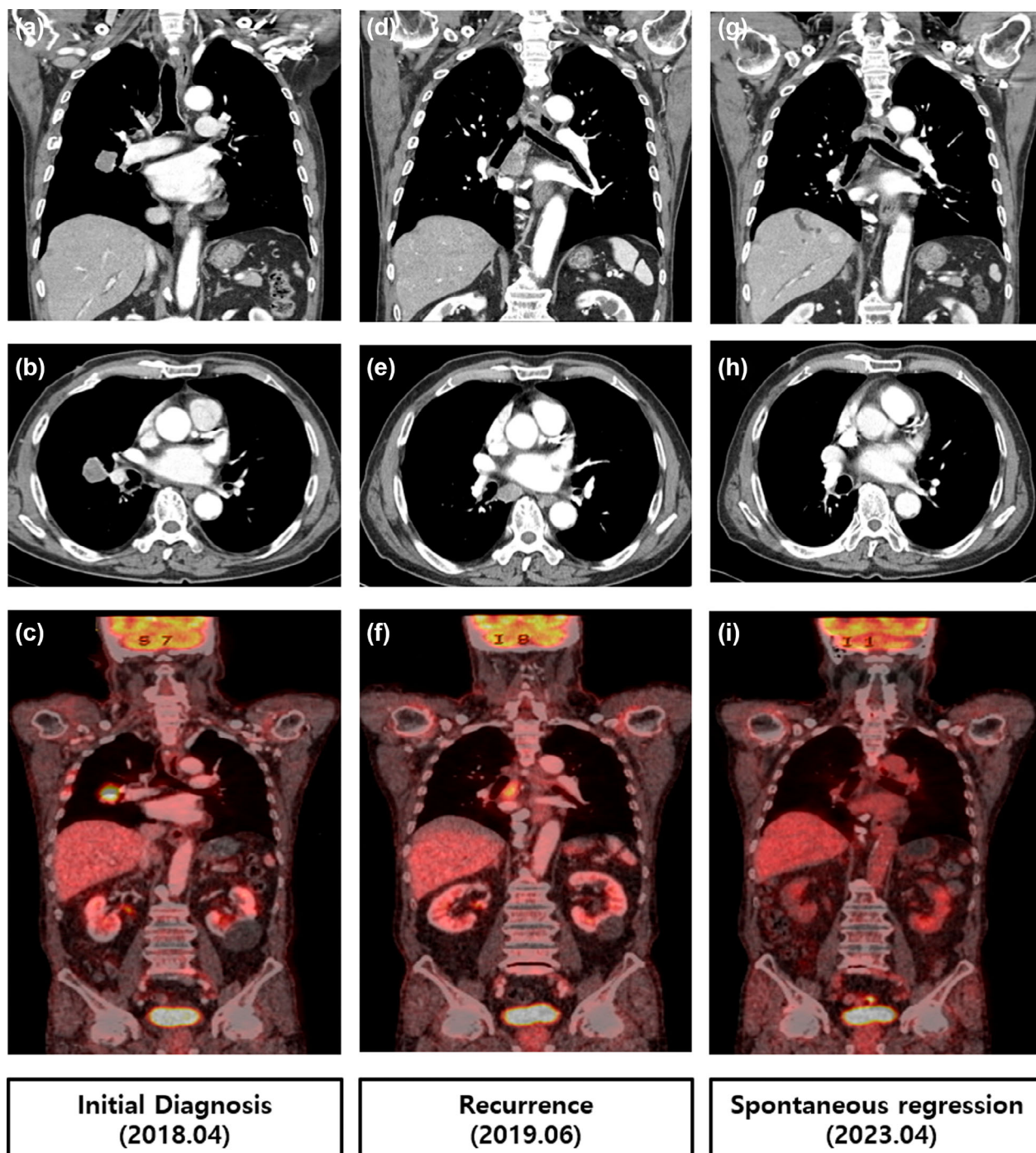


FIGURE 1 Imaging findings and clinical course of the case. Initial diagnosis at imaging: chest computed tomography (CT) scan in (a) coronal view and (b) horizontal view, and (c) F-18 fluorodeoxyglucose-positron emission tomography (FDG-PET) scan. Images obtained 1 year after right middle lobe lobectomy: chest CT scan in (d) coronal view and (e) horizontal view as well as (f) FDG-PET scan revealing a recurrent lesion near the stump site and enlargement of lymph nodes 7 and 4R. Images acquired 5 years after the initial diagnosis: chest CT scan in (g) coronal view and (h) horizontal view, and (i) FDG-PET scan, demonstrating spontaneous regression.

3.2 × 2.5-cm large-cell neuroendocrine carcinoma (Figure 2a), with no lymph node metastasis, resulting in a diagnosis of pathological stage T2aN0. Immunohistochemical staining for Ki-67 showed a positivity rate of 50% (Figure 2b); all the cells showed positive staining for CD56 (Figure 2c). Programmed death-ligand 1 (PD-L1)

immunostaining (Ventana PD-L1 SP263 assay; Roche Diagnostics) showed high PD-L1 expression, with a tumor proportion score of >90% (Figure 2d).

Follow-up chest CT and PET-CT 1 year after surgery showed a new soft-tissue lesion measuring 11 mm next to the stump site and enlargement of lymph nodes 7 and 4R

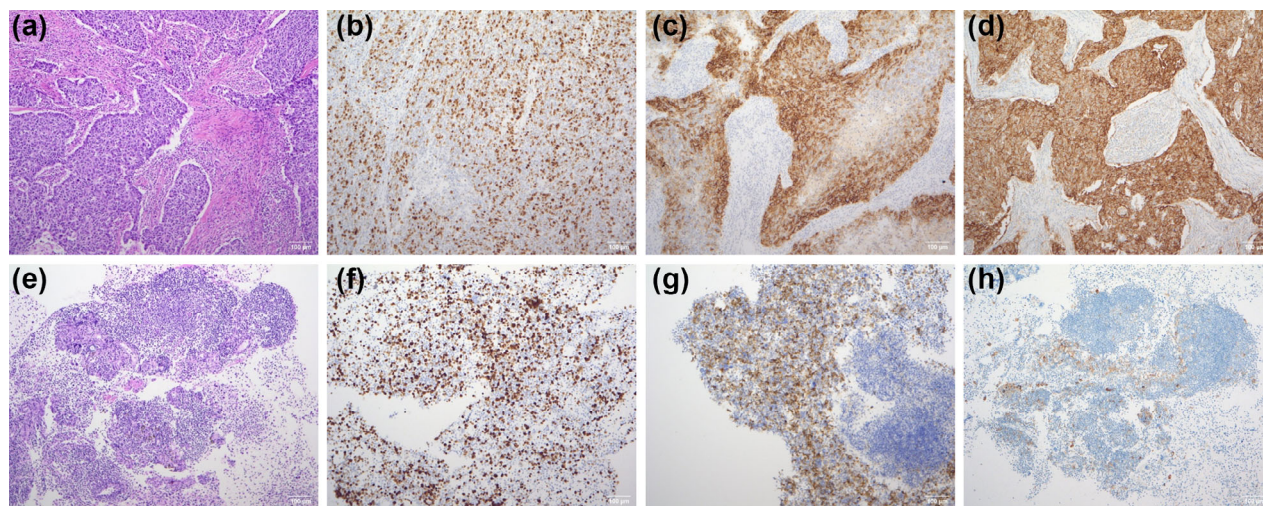


FIGURE 2 Changes in histological features and PD-L1 expression. Histological features revealed on routine hematoxylin and eosin (H&E) staining, showing large cell neuroendocrine carcinoma (LCNEC) at (a) initial diagnosis and (e) at recurrence. (b) Ki-67 immunohistochemical staining of cancer cells at initial diagnosis and (f) recurrence. Immunohistochemical staining demonstrating CD-56 positivity of cancer cells at (c) initial diagnosis and (g) at recurrence. PD-L1 expression levels, with high expression at (d) initial diagnosis and (h) a change to <5% at recurrence.

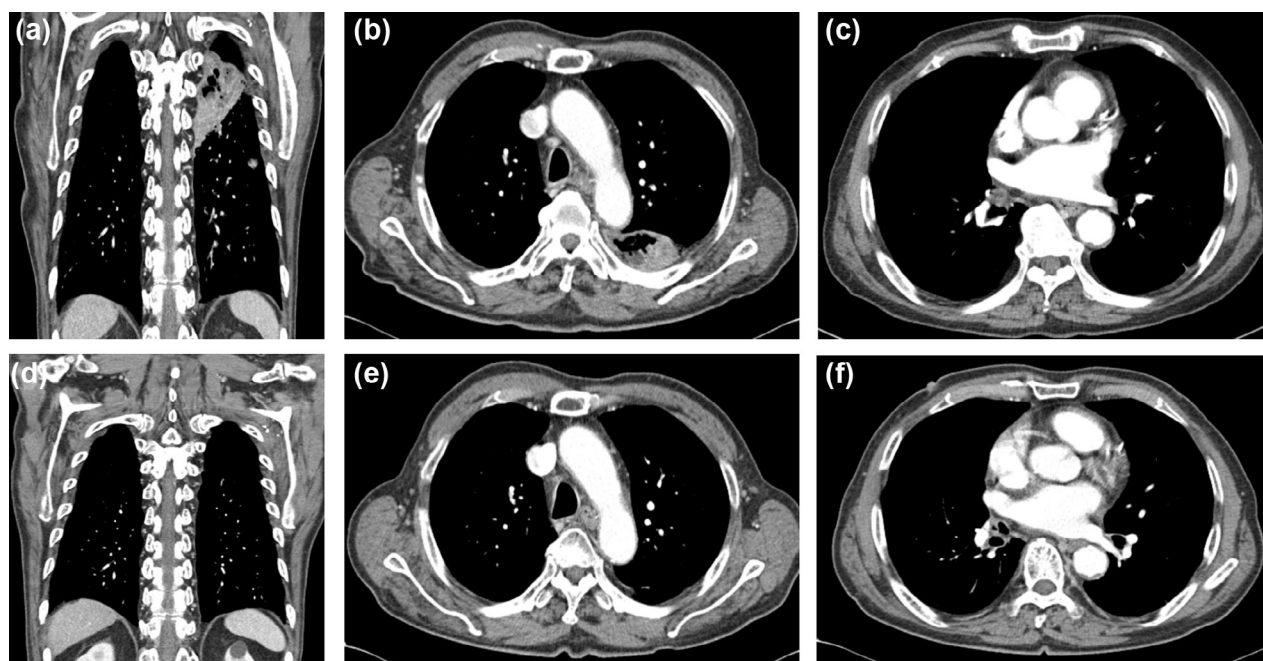


FIGURE 3 Chest computed tomography (CT) images. Images obtained during hospitalization for necrotizing pneumonia of (a) the coronal view, (b) horizontal view of necrotizing pneumonia, and (c) image indicating a recurrent lesion adjacent to the stump site. Chest CT images at the resolution phase of necrotizing pneumonia after 6 months, including (d) the coronal view, (e) horizontal view showing the resolving state, and (f) demonstrating reduction in size of the recurrent lesion.

(Figure 1d–f). Endobronchial ultrasound-guided transbronchial needle aspiration revealed metastatic neuroendocrine carcinoma (Figure 2e) in both lymph nodes. Immunohistochemical staining for CD56 (Figure 2g) and CK yielded positive results, Ki-67 expression was >80% (Figure 2f), and PD-L1 expression was 5% (Figure 2h). The patient refused concurrent chemoradiation therapy for personal reasons and was lost to follow-up. One year after the diagnosis of

recurrence, he visited the emergency room with a fever, cough, and sputum. The blood tests revealed white blood cell (WBC) count of 17 300/ μ L, indicating leukocytosis. The C-reactive protein (CRP) level was elevated at 25.2 mg/dL, and procalcitonin level was 0.38 ng/mL. Multiple nodules with central necrosis in the left upper, left lower, and right lower lobes were observed on chest CT, suggesting multifocal necrotizing pneumonia (Figure 3a–c). The patient was

treated with intravenous antibiotics for 3 weeks, and percutaneous catheter drainage and urokinase instillation were performed for left loculated parapneumonic effusion. Follow-up chest CT (Figure 3d–f) showed improvement in the pneumonia and a gradual decrease in the recurrent lesion, which was not visible after 1 year. No hypermetabolic lesions were observed on PET-CT performed 5 years after the diagnosis of lung cancer, indicating complete spontaneous regression (Figure 1g–i).

DISCUSSION

Spontaneous regression of cancer is a rare phenomenon.⁵ One case report documented spontaneous partial regression of an LCNEC, from 2.8 to 1.8 cm over 4 months. However, 1 year later, the patient began steroid treatment for rheumatoid arthritis, and 6 months later, the tumor size increased to 8.6 cm.⁶ In our patient, spontaneous regression occurred after necrotizing pneumonia, and complete regression occurred at 5 years after the initial diagnosis. Furthermore, in the aforementioned case, the regressed cancer progressed after the use of steroids, whereas in our patient, spontaneous regression occurred after severe pneumonia.

Various theories have been proposed to explain the mechanisms underlying spontaneous cancer regression, including modulated immunological responses following systemic infection and hormonal and psychoneuroimmunological mechanisms.⁷ Various factors can stimulate the immune process such as bacterial products, enzymes, infections, hormones, and trauma.⁵ Recent reports have highlighted the spontaneous regression of various cancers, including hematological malignancies such as acute leukemia and lymphoma and solid tumors such as renal cell carcinoma and colorectal cancer, following COVID-19 infection.⁸ The most frequently reported trigger for spontaneous regression is the development of an acute febrile state resulting from either natural or deliberately induced acute infections.⁹ In our patient, severe inflammation associated with necrotizing pneumonia may have triggered a systemic immune response that was involved in spontaneous regression.

Various factors regulate PD-L1 expression in tumor cells.¹⁰ PD-L1 changes throughout the treatment process have been reported.^{11,12} In our patient, PD-L1 expression decreased from >90% to less than 5% after relapse. Multiple cytokines in the tumor microenvironment (TME) induce changes in PD-L1 expression in tumors and/or immune cells through distinct signaling mechanisms.^{13–16} Changes in the TME and immune-related pathways may have altered PD-L1 expression in our patient.

Our patient experienced spontaneous complete regression of recurrent LCNEC after severe pneumonia, highlighting the potential of immune response and TME in spontaneous regression. Further research is required to understand the immunological mechanisms involved in the spontaneous regression of LCNEC.

AUTHOR CONTRIBUTIONS

Conception and design: Green Hong and Da Hyun Kang; Administrative support: Chaek Chung; Provision of study materials or patients: Dongil Park and Da Hyun Kang; Collection and assembly of data: Green Hong and Song-I Lee; Data analysis and interpretation: Green Hong, Jeong Eun Lee, and Da Hyun Kang; Manuscript writing: Green Hong and Da Hyun Kang; Final approval of manuscript: All authors.

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CONFLICT OF INTEREST STATEMENT

None of the authors report any conflict of interest.

DATA AVAILABILITY STATEMENT

All images used in this article have been uploaded as figures.

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